

UNIT - 4(Historical architecture and modern structural design)

Building Planning:

A site plan or a plot plan is a type of drawing used by architects, landscape architects, urban planners, and engineers which shows existing and proposed conditions for a given area, typically a parcel of land which is to be modified. Sites plan typically show buildings, roads, sidewalks and paths/trails, parking, drainage facilities, sanitary sewer lines, water lines, lighting, and landscaping and garden elements. The orientation of the building is the term used to define the setting or fixing the direction of the layout plan of building respect to the direction of north.

The orientation of building refers to the direction of normal to the normal to the long axis.

For example, if the length of building East-West orientation North-South.

Good orientation means proper placement of room in relation providing convenient access both to the street and back-yard

- Top 10 most common types of Construction drawings use regularly in construction industries.

1. Block Plan

This drawing gives a layout of the site or the buildings in the surrounding area, laid out on a map drawn to scale.

It gives a first-hand idea of the roads, boundaries and other such details that are necessary to understand where your construction site lies. It helps the person dealing with your construction plan or project request to understand what and where you are proposing it and help you out with it too.

Block plans are made in relation to Ordnance Survey Maps and the recommended scales used are 1:2500, 1:1250 or 1:500.

2. Architectural Drawings

Architectural Drawings are drawing work that is used in building drawings to depict the dimensions, depth, and layout of the actual building, prior to beginning the construction. Architectural Drawings act as a blueprint construction, drawn to scale, to help the engineers visualize the project.

Various types of Architectural Drawings commonly used are:

- Foundation plan**
- Floor plans**
- Sectional Drawings**
- Elevation Drawings**

3. Production Drawings

These Construction Drawings are used to convey functional information to the workers and engineers on site. It describes the materials, the assembly of various parts, the tools required, the dimensions, and other information required during the process. It may also include additional information or an infographic on how to meet those set requirements.

4. Structural Drawings

Structural Drawings also serve as civil engineering drawings. They are useful in understanding the physical nitty-gritty of a building framework. They act as a structural design guide for the workers and on-site engineers. Common types of structural drawings are:

General Note – an overview of all the codes, procedures, and abbreviations, etc required to give a comprehensive guide to getting to work on the construction site. This includes concrete mix, details for other structural drawings, lengths and construction types of each component, etc.

5. Electrical drawings

Most residential construction drawings or commercial construction drawings require a functional outline of the number of power outlets, light fixtures, fan fixtures, etc. They also include the wiring pattern and details about the electrical load it can carry. Common details included in Electrical Drawings are:

- ☐ **Earthing layout**
- ☐ **Light fixture layout**
- ☐ **Generator and other equipment**
- ☐ **Cable tray layout**
- ☐ **Hazardous area classifications**
- ☐ **Lighting protection system**

6. Plumbing Drawings

Just like electrical layouts, plumbing is another part of any residential or commercial construction drawing that marks the points where plumbing components need to be set up.

Space is left here accordingly for further pipe and sanitary ware fixtures to be added once the structural component is finished. Plumbing drawings commonly include:

- Pipes -water pipes, drainage pipes, internal pipes
- Material of pipes
- Outlet points - taps, sinks, tanks etc

7. HVAC Drawings

These are known as mechanical construction drawings. They provide details and a design framework for heating and ventilation systems in a building. Central heating/cooling, air conditioning vents, ventilators, etc are all included according to the need and site of the building plans. Builders use these design constructs in their process accordingly.

8. Firefighting Drawings

In today's construction systems, safety design is paramount. Firefight Drawings are also a part of blueprint drawings of a building that allocate points for fire hoses, fire escapes, water outlets, sandbags, or any other fire safety equipment required by the regulatory body overseeing the project.

9. Environmental Plans

Making sure environmental guidelines and management is properly followed is a part of projects that cannot be overlooked. The aim is to minimize environmental damage and future negative impacts of the construction project.

10. Finishing Drawings

These include finer and more detailed plans of the building after the whole structural and architectural framework has been set up. These are required for the aesthetic and functional value of the building. These construction drawings include details of:

Tile patterns,

- Floor patterns
- False ceilings
- Paint colours and textures
- Plaster
- Woodwork

Standard codes as measure of controlling safety and serviceability

It is not only necessary but rather indispensable for civil engineers and professionals engaged in civil engineering practices to have a practical understanding of the modification and changes in the Indian standard codes of practice taking place time to time by the Bureau of Indian Standards, New Delhi. Major changes such as Limit State Design method, change in seismic zones, new load considerations, mass irregularities, dynamic loading analysis, etc. are to be followed so that safe structures with long serviceability can be accomplished. The dynamics of change and regular adjustment in IS codes cannot be ignored if at all the structures built

Standard codes of practice for civil engineering enumerate prospects and necessity of the design and construction practices of the structures. They provide a set of rules and regulations that specify the standards for the design of all types of structures within the ambit of civil engineering. The purpose of codes is to provide minimum guiding standards for safety, health and general civil works (encompassing sanitation, sewerage, water supply, water drainage, rain water harvesting, lighting and ventilation), structural and mechanical integrity, means of egress, fire protection & control and energy conservation

Limit State of Collapse

- (i) Limit state of collapse in flexure (bending)
- (ii) Limit state of collapse in compression
- (iii) Limit state of collapse in shear
- (iv) Limit state of collapse in torsion.

Limit State of Serviceability

- (i) Limit state of deflection
- (ii) Limit state of cracking
- (iii) Limit state of vibration

The codes which follow limit state of collapse and serviceability are -

IS 456 : 2000 , ACI 318-11 , EN 1990 - EN 1999
Building bye-laws' purpose

Formulated primarily by a central authority, building bye-laws ensure that constructions are not only safe but also adhere to aesthetic standards. In that sense, these regulate the construction and the architectural aspects of construction activities. For example, the rules prescribed under the building bye-laws can make it mandatory for builders to keep fire safety and earthquake-resistance provisions at their projects. Building bye-laws also govern the provisions for open spaces in a project, with the aim to ensure that developments do not turn the city into a concrete jungle.

What aspects do building bye-laws cover?

Under the building bye-laws in India, directions are laid down with respect to the following aspects of construction:

1. Area and usage
2. Building height
3. Building coverage
4. Floor space index
5. Density

6. Setbacks and projections

7. Parking facilities

8. Fire provisions with respect to staircase and exits

9. Basement facilities

10. Green spaces

11. Open spaces

12. Amenities in the project

13. Provision for elevators

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- 14.Sewerage facilities
 - 15.Provision for water
 - 16.Provision for power supply
 - 17.Provision for waste management
 - 18.Rainwater harvesting
 - 19.Barrier-free environment
 - 20.Safety provisions
 - 21.Effects of communication technology

Concept of Green Buildings

A 'green' building is a building that, in its design, construction or operation, reduces or eliminates negative impacts, and can create positive impacts, on our climate and natural environment. Green buildings preserve precious natural resources and improve our quality of life.

There are a number of features which can make a building 'green'. These include:

- Efficient use of energy, water and other resources
- Use of renewable energy, such as solar energy
- Pollution and waste reduction measures, and the enabling of re-use and recycling
- Good indoor environmental air quality
- Use of materials that are non-toxic, ethical and sustainable
- Consideration of the environment in design, construction and operation
- Consideration of the quality of life of occupants in design, construction and operation
- A design that enables adaptation to a changing

Constructions:

1. Masonry

Masonry is the building of structures from individual units, which are often laid in and bound together by mortar; the term masonry can also refer to the units themselves. The common materials of masonry construction are brick, building stone such as marble, granite, and limestone, cast stone, concrete block, glass block, and adobe. Masonry is generally a highly durable form of construction.

Applications

Masonry is commonly used for walls and buildings. Brick and concrete block are the most common types of masonry in use in industrialized nations and may be either load-bearing or non-load-bearing.

Advantages

The use of materials such as bricks and stones can increase the thermal mass of a building. Masonry is a non-combustible product and can protect the building from fire. Masonry walls are more resistant to projectiles, such as debris from hurricanes or tornadoes.

2. Timber

Timber is a type of wood which has been processed into beams and planks. It is also known as “**lumber**” in US and Canada. Basically, timber or Lumber is a wood or firewood of growing trees. Any wood capable of yielding a minimum dimensional size can be termed as a timber or lumber. It is a stage in the process of wood production. Timbers are used for the structural purpose. Those woods which are adapted for building purposes are timbers. Finished timber is supplied in standard sizes for the industry. Timber is used for building houses and making furniture.

Steel and concrete constructions

Concrete and steel are among the most common construction materials, and many building projects can use either of them for their underlying structure. Each material has advantages and limitations, like in any engineering decision, and this article provides a general comparison. No material can be considered better than the other for all cases, and the best option is determined by project conditions.